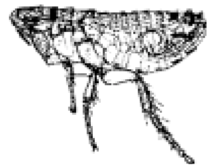
flea



NEWS 82 *June 2018*

http://esanetworks.org/

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Contents

Editorial

Dear *Flea News* Reader,

New research on the Black Death plagues of the Middle Ages blames human ectoparasites as the essential vectors. See the abstracts in the previous Flea News 82 (Dean et al. 2017, Ditrich 2017).

Rats (*Rattus rattus* and *R. norvegicus*) get sick with plague and have long been implicated, but their populations did not die-off as expected and the disease spread too rapidly for rats to cause the outbreaks, according to the new research using historical records.

Some human ectoparasites can carry plague. Fleas (human flea, *Pulex irritans*, and Oriental rat flea, *Xenopsylla cheopis*) that are infected can develop a gut blockage that may enhance their capacity for transmitting plague. Human lice (human body lice, *Pediculus humanus*) are infected naturally with plague, too, and transmit plague experimentally, but their vectorial capacity is unclear, and considered low (human pubic lice, *Pthirus pubis*, are considered unimportant in plague).

Humans themselves, or their cats and dogs, can be sources of plague, especially as a respiratory infection. Pertinent is the hypothesis that plague bacteria, *Yersinia pestis*, evolved from *Y. pseudotuberculosis*.

Complicating our understanding is the question: to what extent is modern plague different from ancient plague? Evolution of the bacterium and its vectors, as well as changes in cultural factors such as hygiene, could have changed the disease. The transmission cycle of plague, medieval and modern, is still controversial.

Yours in fleas,
R.L. Bossard, Editor, *Flea News*

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Katharine R. Dean, Fabienne Krauer, Lars Walløe, Ole Christian Lingjærde, Barbara Bramanti, Nils

Chr. Stenseth, Boris V. Schmid. 2018. Human ectoparasites and spread of plague in Europe. Proceedings of the National Academy of Sciences Jan 2018, 201715640; DOI: 10.1073/pnas.1715640115

Ditrich, H. (2017) The transmission of the Black Death to Western Europe: a critical review of the existing evidence. Mediterranean Historical Review, 32 (1), pp. 25-39.

Linda Houhamdi, Hubert Lepidi, Michel Drancourt, Didier Raoult; Experimental Model to Evaluate the Human Body Louse as a Vector of Plague. The Journal of Infectious Diseases, Volume 194, Issue 11, 1 December 2006, Pages 1589–1596, https://doi.org/10.1086/508995

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Spyrou, M. A., R. I. Tukhbatova, C. -C Wang, A. A. Valtueña, A. K. Lankapalli, V. V. Kondrashin, V. A. Tsybin, et al. 2018. "Analysis of 3800-Year-Old *Yersinia pestis* Genomes Suggests Bronze Age Origin for Bubonic Plague." Nature Communications 9 (1). doi:10.1038/s41467-018-04550-9.

Walløe, L. (2008). 3. Medieval and Modern Bubonic Plague: Some Clinical Continuities. Medical History, 52(S27), 59-73.

Welford, M. 2018. "Geographies of Plague Pandemics: The Spatial-Temporal Behavior of Plague to the Modern Day." 1-156. doi:10.4324/9781315307435.

Announcements

Flea Bites Increasing

The Centers for Disease Control (CDC) estimates "Disease cases from mosquito, tick, and flea bites tripled in the US from 2004 to 2016...", and "Nine new germs spread by mosquitoes and ticks have been discovered or introduced since 2004."

Vital Signs MAY 2018, www.cdc.gov/vitalsigns/vector-borne

Congratulations to Christine M. McCoy on her new position as Staff Entomologist for BerTek.

The 2018 Livestock Insect Workers Conference was held June 17-20 in San Juan, Puerto Rico.

Next summer (July 7-11, 2019) our Livestock Insect Workers Conference will be held in conjunction with the AAVP and the WAAVP in Madison, Wisconsin http://www.waavp2019.com/ (AMERICAN ASSOCIATION OF VETERINARY PARASITOLOGISTS and the WORLD ASSOCIATION FOR THE ADVANCEMENT OF VETERINARY PARASITOLOGY).

Thanks to everyone sending corrections to the Updated Listing of Flea Species of the World.
R.L. Bossard, Editor, *Flea News*

Featured Research

Flea Taxonomy

Zurita, A., R. Callejón, M. de Rojas, and C. Cutillas. 2017. "Morphological, Biometrical and Molecular Characterization of *Archaeopsylla erinacei* (Bouché, 1835)." Bulletin of Entomological Research: 1-13. doi:10.1017/S0007485317001274.

In the present work, we carried out a morphological, biometrical and molecular study of the species *Archaeopsylla erinacei* (Bouché, 1835) and their subspecies: *Archaeopsylla erinacei erinacei* (Bouché, 1835) and *Archaeopsylla erinacei maura* (Jordan & Rothschild, 1912) isolated from hedgehogs (*Erinaceus europaeus*) from different geographical regions (Seville and Corse). We have found morphological differences in females of *A. erinacei* from the same geographical origin that did not correspond with molecular differences. We suggest that some morphological characters traditionally used to discriminate females of both subspecies should be revised as well as we set the total length of the spermatheca as a valid criterion in order to discriminate between both subspecies. The Internal Transcribed Spacers 1 and 2 (ITS1, ITS2) and partial 18S rRNA gene, and partial cytochrome c-oxidase 1 (cox1) and cytochrome b (cytb) mtDNA gene sequences were determined to clarify the taxonomic status of these taxa and to assess intra-specific and intra-population similarity. In addition, a phylogenetic analysis with other species of fleas using Bayesian and Maximum Likelihood analysis was performed. All molecular markers used, except 18S, showed molecular differences between populations corresponding with geographical origins. Thus, based on the phylogenetic and molecular study of two nuclear markers (ITS1, ITS2) and two mitochondrial markers (cox1 and cytb), as well as concatenated sequences of both subspecies, we reported the existence of two geographical genetic lineages in *A. erinacei* corresponding with two different subspecies: *A. e. erinacei* (Corse, France) and *A. e. maura* (Seville, Spain), that could be discriminated by polymerase chain reaction-linked random-fragment-length polymorphism.

Hosts and Distribution

Beaucournu J.-C. 2017. Puces collectées dans deux grottes de Cazorla, Province de Jaen (Espagne) : grottes Navilla de Fuente Acera n°1 et 2. *Monografias Biospeleogicas* 12 : 6-8.

Keskin A., Hastriter M.W. & Beaucournu J.-C. 2018. Fleas (Insecta) of Turkey : species composition, geographical distribution and host associations. *Zootaxa* 4420 (2) : 211-228.

A list of the fleas species reported from Turkey is provided, with their geographical distribution and host associations. A total of 115 flea taxa (83 species, 32 subspecies) belonging to 36 genera and seven families are listed. The most common families are Ctenophthalmidae (n= 44) and

Ceratophyllidae (27), followed by Ischnopsyllidae (13), Leptopsyllidae (11), Pulicidae (11), Vermipsyllidae (5) and Hystrichopsyllidae (4).

Lareschi, M. Description of the males of *Androlaelaps misionalis* and *Androlaelaps ulysespardinasi* (Acari: Parasitiformes: Laelapidae) parasitic of sigmondontine rodents from northeastern Argentina. Journal of Parasitology. In press. https://doi.org /10.1645/18-1

Lareschi, M., JM Venzal, S Nava, AJ Mangold, A Portillo, AM Palomar Urbina, & JA Oteo Revuelta. 2018. The human flea *Pulex irritans* Linnaeus, 1758 (Siphonaptera: Pulicidae) and an investigation of *Bartonella* and *Rickettsia* in northwestern Argentina. Revista Mexicana de Biodiversidad 89: 375-381.

Pulex irritans is the only cosmopolitan flea species and the most studied one within the genus *Pulex*. It has importance in public health since it commonly parasitizes humans causing dermatitis, and it has been also implicated in the transmission of bacterial pathogens. *Pulex irritans* has been confused with the closely related *Pulex simulans* species for years. Herein, *Pulex* specimens collected from a Pampas fox and a Chacoan peccary from northwestern Argentina were identified by comparison with type specimens. In addition, the presence of *Bartonella* spp. and *Rickettsia* spp. was investigated using PCR assays. Our results provided characters of diagnostic importance to identify *P. irritans*, which include the shape of sternite VII in the females, and of the aedeagal sclerite, clasper and crochet in the males. Besides, we report for the first time *P. irritans* parasitizing a peccary. This finding reinforces the hypothesis of the origin of this flea associated with this mammal, and then colonizing humans and domestic mammals. There was no evidence of *Bartonella* or *Rickettsia* DNA in the analyzed fleas. This information even if negative may be considered relevant for *P. irritans* from Argentina.

Sanchez, J, Lareschi, M. 2018. Ecological approach of the fleas associated with Sigmodontine rodents from Patagonia, Argentina. Bulletin of Entomological Research, 1–12. In press doi:10.1017/S0007485318000196

Fleas have great medical relevance as vectors of the causative agents of several diseases in animals and humans and rodents are the principal reservoirs for these pathogens. Argentinian Patagonia has the highest diversity of rodent fleas in South America. However, parasitism rates of rodents by fleas, the factors that influence them and the ecological aspects that modulate geographical distributions of flea–host association remain unknown for this region. This is the first study to record the diversity, prevalence, abundance, geographical distributions and host ranges of fleas in Argentinian Patagonia. It also compares parasitism rates among Patagonian ecoregions and host species. We captured 438 rodents belonging to 13 species, which harboured 624 fleas from 11 species and subspecies (P = 46%; mean abundance = 1.44). The high parasitism rates obtained were consistent with previous records for other arid regions, suggesting that Patagonia favours the survival and development of Siphonaptera. Host geographic range and abundance were related to the parasitological indexes: host species with high-density populations had the highest mean flea abundance and prevalence, whereas widely distributed hosts had the highest richness and diversity of flea species. Our results contribute to the knowledge of the flea–host–environment complex. Our analysis of flea distributions and parasitism rate in Central Patagonia may be useful in epidemiological studies of flea-borne diseases and provide a basis for implementing surveillance systems for better risk assessment of emerging zoonoses in the region.

Sanchez, J., A. Travaini, A. Rodriguez, & M. Lareschi. 2018. Primer estudio de los índices parasitológicos de dos especies de pulgas (Insecta, Siphonaptera) parásitas de zorros de la Patagonia Argentina: *Pulex irritans* (Pulicidae) y *Polygenis (P.) platensis* (Rhopalopsyllidae). Mastozoología Neotropical. In press

López-Berrizbeitia, M. F., J. Sanchez, R. M. Barquez, and M. M. Díaz. 2018. "Descriptions of Two New Species of Flea of the Genus *Plocopsylla* in Northwestern Argentina." *Medical and Veterinary Entomology*. doi:10.1111/mve.12303.

Liljesthrom, G, Lareschi M. Predicting species richness of ectoparasites of wild rodents from the Río de la Plata coastal wetlands, Argentina. Parasitology Research. In press.

Sanchez, J., M. Lareschi, J. Salazar-Bravo, & S. L. Gardner. Fleas of genus *Neotyphloceras* associated with rodents from Bolivia: New host and distributional records, description of a new species and remarks on the morphology of *Neotyphloceras rosenbergi*. Medical and Veterinary Entomology. In press

SANCHEZ J.P., EZQUIAGA, M.C. & M. RUIZ. 2018. Fleas (Insecta: Siphonaptera) with public health relevance in domestic pigs (Artiodactyla: Suidae) from Argentina. Zootaxa, 4374 (1):144-150.

SANCHEZ J.P. & Y. GUROVICH. Fleas (Insecta: Siphonaptera) associated to the endangered Neotropical marsupial Monito del Monte (*Dromiciops gliroides* Microbiotheria: Microbiotheriidae) in its Southernmost Population of Argentina. Mastozoología Neotropical (In Press)

Harimalala, M., Miarinjara, A., Duchemin, J.-B., Ramihangihajason, T., Boyer, S. Species diversity and phylogeny of fleas of small terrestrial mammals in the forests of the Central Highlands of Madagascar. (2018) Zootaxa, 4399 (2), pp. 181-196.

Fleas are holometabolous insects forming the order of Siphonaptera. Some studies have been carried out on biology and systematic of Malagasy fleas, but little is known about their phylogenetic relationships. In this study, we focused on flea species occurring in the forests of the Central Highlands and also, on the determination of their phylogenetic relationships. Three families, five genera and thirteen species were identified. The family Pulicidae includes four species (*Centetipsylla madagascariensis* Rothschild, *Synopsyllus fonquerniei* Wagner & Roubaud, *S. estradei* Klein and *S. robici* Klein); Leptopsyllidae has eight species (*Paractenopsyllus vauceli* Klein, *P. petiti* Klein, *P. viettei* Klein, *P. grandidieri* Klein, *P. goodmani* Duchemin, *P. rouxi* Duchemin, *P. raxworthyi*

Duchemin & Ratovonjato and *Tsaractenus rodhaini* Duchemin), and Ctenophtalmidae one species (*Dinopsyllus brachypecten* Smit). All are endemic to Madagascar and each differs geographically. Flea phylogenetic relationships were inferred using four molecular markers (ITS2, mtCOII, 16SrRNA and 12S rR-NA) and using Neighbor-Joining, Maximum Parsimony and Bayesian methods with addition of Genbank sequences of exotic species. The Family Pulicidae was monophyletic while the families Leptopsyllidae and Ctenophtalmidae were paraphyletic. Malagasy fleas are homogeneous and all species adhere to current classification schemes.

Ecology

OBERTO, V. S. C., GULARTE, F. D. A., RODRIGUES, J. D. F., & DA SILVA, G. A. C. (2018). Avaliação de dietas para criação de *Ctenocephalides felis* in vitro. ANAIS DA 14ª MOSTRA DE INICIAÇÃO CIENTÍFICA-CONGREGA URCAMP-2017, 439-440. [Evaluation of diets for rearing *Ctenocephalides felis* in vitro]

Introdução: A pulga *Ctenocephalides felis* é um ectoparasita de importância veterinária e para saúde pública. Este sifonáptero, através da hematofagia, pode ser vetor de helmintos, cestódeos e filarídeos, assim como outros patógenos, como bactérias, rickettsias e vírus. Estapulga tem distribuição mundial, sendo um parasita de várias espécies de animais domésticos e silvestres. Em cães e em gatos domésticos, é o ectoparasita mais prevalente, sendo seu parasitismo associado à ação irritativa e espoliadora e, eventualmente, à dematite alérgica.

Objetivo: O objetivo do presente estudo foi determinar o efeito nutritivo de quatro diferentes dietas sobre a emergência de adultos *C. felis*.

Materiais e Métodos: As dietas artificiais foram compostas por farinha de sangue bovino e eqüino adicionadas a farelo in natura de arroz ou de trigo, além de areia peneirada. Dois cães naturalmente infestados com *C. felis* foram mantidos por doze (12) horas em gaiolas metálicas com bandejas plásticas abaixo para coleta de ovos, sendo estes coletados com pincel de cerda única. As dietas foram compostas por sangue eqüino e bovino desidratado em estufa a 100ºC por 24 horas, e farelo in natura de arroz e de trigo e areia fina peneirada. Em cada frasco de vidro, vedado com tela de nylon, foram colocados vinte (20) ovos de pulgas e acrescentada a dieta, composta por 2,0 gramas de uma mistura a uma proporção 1:1 de farelo/farinha juntamente a 0,4g da areia, com cinco repetições para cada dieta. Os frascos foram incubados em câmara climatizada (B.O.D) na temperatura de 27 ± 1ºC e 75 ± 10% de umidade relativa por um período de trinta (30) dias, quando seu conteúdo foi fixado em álcool 70% e separado em placas de Petri, separando-se os adultos eclodidos e as demais formas evolutivas que não completaram o ciclo.

Resultados: Na dieta composta por sangue equino e farelo de trigo, coletaram-se 32 espécimes adultos (32% de eclosão), sendo 20 fêmeas e 12 machos. Na dieta de bovino e trigo foram recuperados 9 adultos (9% de eclosão), sendo 6 fêmeas e 3 machos, além de 11 formas imaturas que não completaram o ciclo. Na dieta bovino e arroz, recuperou-se apenas um macho adulto (1% de eclosão). Na dieta composta por sangue equino e farelo de arroz, não houve eclosão. Sugere-se que a maior recuperação de adultos nas dietas com sangue equino deu-se ao fato de esta espécie possuir um maior teor de hemoglobina, o que é fundamental para o desenvolvimento do parasita. Com relação ao tipo de farelo, o farelo de arroz apresentou maior umidade, o que pode ter favorecido o crescimento fúngico.

Conclusão: Conclui-se que, dentre as dietas experimentadas, a que apresentou maiores percentuais de recuperação de adultos foi a composta por farelo de trigo e sangue equino. Mais estudos devem ser aplicados no intuito de se estabelecer a padronização de uma dieta para criação de *C. felis*, permitindo a manutenção de colônias e consequentes estudos com esses insetos.

[Objective: The objective of the present study was to determine the nutritive effect of four different diets on the emergence of *C. felis* adults...

Results: In the diet composed of equine blood and wheat bran, 32 adult specimens (32% of hatching) were collected, 20 females and 12 males. In the bovine and wheat diet, 9 adults (9% of hatching) were recovered, 6 females and 3 males, in addition to 11 immature forms that did not complete the cycle. In the bovine and rice diet, only one adult male (1% hatching) was recovered. In the diet composed of equine blood and rice bran, there was no hatching. It is suggested that the greater recovery of adults in the diets with equine blood was due to the fact that this species has a higher content of hemoglobin, which is fundamental for the development of the parasite. Regarding the type of bran, rice bran presented higher humidity, which may have favored fungal growth.

Conclusion: It was concluded that, among the diets tested, the one with the highest percentages of adult recovery was composed of wheat bran and equine blood. Further studies should be applied in order to establish the standardization of a diet for the creation of *C. felis*, allowing colonies to be maintained and consequent studies with these insects.]

Bush, S.E., Clayton, D.H. (2018) Anti-parasite behaviour of birds. Philosophical Transactions of the Royal Society B: Biological Sciences, 373 (1751), art. no. 20170196, 13 p.

Birds have many kinds of internal and external parasites, including viruses, bacteria and fungi, as well as protozoa, helminths and arthropods. Because parasites have negative effects on host fitness, selection favours the evolution of anti-parasite defences, many of which involve behaviour. We provide a brief review of anti-parasite behaviours in birds, divided into five major categories: (i) body maintenance, (ii) nest maintenance, (iii) avoidance of parasitized prey, (iv) migration and (v) tolerance. We evaluate the adaptive significance of the different behaviours and note cases in which additional research is particularly needed. We briefly consider the interaction of different behaviours, such as sunning and preening, and how behavioural defences may interact with other forms of defence, such as immune responses. We conclude by suggesting some general questions that need to be addressed concerning the nature of anti-parasite behaviour in birds. This article is part of the Theo Murphy meeting issue ‘Evolution of pathogen and parasite avoidance behaviours’.

van der Mescht, L., Khokhlova, I.S., Warburton, E.M., Dlugosz, E.M., Kotler, B.P., Krasnov, B.R.

(2018) Can we predict the success of a parasite to colonise an invasive host? Parasitology Research, 117 (7), pp. 2305-2314.

To understand whether a parasite can exploit a novel invasive host species, we measured reproductive performance (number of eggs per female per day, egg size, development rate and size of new imagoes) of fleas from the Negev desert in Israel (two host generalists, *Synosternus cleopatrae* and *Xenopsylla ramesis*, and a host specialist, *Parapulex chephrenis*) when they exploited either a local murid host (*Gerbillus andersoni*, *Meriones crassus* and *Acomys cahirinus*) or two alien hosts (North American heteromyids, *Chaetodipus penicillatus* and *Dipodomys merriami*). We asked whether (1) reproductive performance of a flea differs between an alien and a characteristic hosts and (2) this difference is greater in a host specialist than in host generalists. The three fleas performed poorly on alien hosts as compared to local hosts, but the pattern of performance differed both among fleas and within fleas between alien hosts. The response to alien hosts did not depend on the degree of host specificity of a flea. We conclude that successful parasite colonisation of an invasive host is determined by some physiological, immunological and/or behavioural compatibility between a host and a parasite. This compatibility is unique for each host-parasite association, so that the success of a parasite to colonise an invasive host is unpredictable.

Young, T.P., Porensky, L.M., Riginos, C., Veblen, K.E., Odadi, W.O., Kimuyu, D.M., Charles, G.K., Young, H.S.(2018) Relationships Between Cattle and Biodiversity in Multiuse Landscape Revealed by Kenya Long-Term Exclosure Experiment. Rangeland Ecology and Management, 71 (3), pp. 281-291.

On rangelands worldwide, cattle interact with many forms of biodiversity, most obviously with vegetation and other large herbivores. Since 1995, we have been manipulating the presence of cattle, mesoherbivores, and megaherbivores (elephants and giraffes) in a series of eighteen 4-ha (10-acre) plots at the Kenya Long-term Exclosure Experiment. We recently (2013) crossed these treatments with small-scale controlled burns. These replicated experimental treatments simulate different land management practices. We seek to disentangle the complex relationships between livestock and biodiversity in a biome where worldwide, uneasy coexistence is the norm. Here, we synthesize more than 20 yr of data to address three central questions about the potentially unique role of cattle in savanna ecology: 1) To what extent do cattle and wild herbivores compete with or facilitate each other? 2) Are the effects of cattle on vegetation similar to those of wildlife, or do cattle have unique effects? 3) What effects do cattle and commercial cattle management have on other savanna organisms? We found that 1) Cattle compete at least as strongly with browsers as grazers, and wildlife compete with cattle, although these negative effects are mitigated by cryptic herbivores (rodents), rainfall, fire, and elephants. 2) Cattle effects on herbaceous vegetation (composition, productivity) are similar to those of the rich mixture of ungulates they replace, differing mainly due to the greater densities of cattle. In contrast, cattle, wild mesoherbivores, and megaherbivores have strongly guild-specific effects on woody vegetation. 3) Both cattle and wild ungulates regulate cascades to other consumers, notably termites, rodents, and disease vectors (ticks and fleas) and pathogens. Overall, cattle management, at moderate stocking densities, can be compatible with the maintenance of considerable native biodiversity, although reducing livestock to these densities in African rangelands is a major challenge.

Cassin Sackett, L. (2018) Does the host matter? Variable influence of host traits on parasitism rates. International Journal for Parasitology, 48 (1), pp. 27-39.

Parasitism of mammals is ubiquitous, but the processes driving parasite aggregation on hosts are poorly understood, as each system seems to show unique correlations between parasitism and host traits such as sex, age, size and body mass. Genetic diversity is also posited to influence susceptibility to parasitism, and provides a quantifiable measure of an intrinsic unchanging host property, but this link has not been well established. A lack of consistency in host traits predicting parasite heterogeneity may derive from the contribution of environmental factors to parasite aggregation. To evaluate this question, a large dataset was leveraged to explore the relationship between unchanging, intrinsic host traits (heterozygosity and sex), variable host traits (age, length and body mass), and extrinsic factors (sampling date/year and population) and flea presence/absence, abundance and intensity on two species of social burrowing mammal, the black-tailed prairie dog (*Cynomys ludovicianus*) and the Gunnison's prairie dog (*Cynomys gunnisoni*). Prairie dogs experience frequent parasitism by fleas, but the distribution of fleas among individuals is highly skewed. In these systems, intrinsic host traits were nuanced in how they predicted flea aggregation on individual prairie dogs, with sex unimportant to parasitism rates and heterozygosity increasing the probability of infection and influencing the number of fleas in divergent ways. Variable host traits interacted with each other and with environmental or geographic stochasticity to influence flea aggregation. Length and age tended to increase parasitism, whereas the effects of body mass and condition were mediated by date and other host traits to produce both positive and negative effects on parasitism. This finding suggests that the factors affecting ectoparasite infection on individuals are complex, even within species. Importantly, there was no correlation between the number of fleas on an individual in one year and the number of fleas on the same individual the next year, supporting the idea that flea aggregation is not driven by unchanging, intrinsic characteristics of the host. Rather, these findings indicate that host traits influence parasitism in nuanced ways, including interactions with environmental characteristics and stochastic factors.

Evolution

Gokhman, V. E. and V. G. Kuznetsova. 2018. "Parthenogenesis in Hexapoda: Holometabolous Insects." *Journal of Zoological Systematics and Evolutionary Research* 56 (1): 23-34. doi:10.1111/jzs.12183.

["Mecoptera (scorpionflies or hanging flies) and Siphonaptera (fleas) Parthenogenesis has never been reported for members of these two smaller insect orders".]

Wood, J.R. (2018) DNA barcoding of ancient parasites. Parasitology, 145 (5), pp. 646-655.

Ancient samples present a number of technical challenges for DNA barcoding, including damaged DNA with low endogenous copy number and short fragment lengths. Nevertheless, techniques are available to overcome these issues, and DNA barcoding has now been used to successfully recover parasite DNA from a wide variety of ancient substrates, including coprolites, cesspit sediment, mummified tissues, burial sediments and permafrost soils. The study of parasite DNA from ancient samples can provide a number of unique scientific insights, for example: (1) into the parasite communities and health of prehistoric human populations; (2) the ability to reconstruct the natural parasite faunas of rare or extinct host species, which has implications for conservation management and de-extinction; and (3) the ability to view in 'real-time' processes that may operate over century- or millennial-timescales, such as how parasites responded to past climate change events or how they co-evolved alongside their hosts. The application of DNA metabarcoding and highthroughput sequencing to ancient specimens has so far been limited, but in future promises great potential for gaining empirical data on poorly understood processes such as parasite co-extinction.

Meng Wang, Bao-Zhen Hua. 2018. A new species of *Neopanorpa* with an extremely long notal organ from Sichuan, China (Mecoptera, Panorpidae). ZooKeys doi: 10.3897/zookeys.750.23486

Appelgren, A.S.C., Saladin, V., Richner, H., Doligez, B., McCoy, K.D.(2018) Gene flow and adaptive potential in a generalist ectoparasite. BMC Evolutionary Biology, 18 (1), art. no. 99.

Background: In host-parasite systems, relative dispersal rates condition genetic novelty within populations and thus their adaptive potential. Knowledge of host and parasite dispersal rates can therefore help us to understand current interaction patterns in wild populations and why these patterns shift over time and space. For generalist parasites however, estimates of dispersal rates depend on both host range and the considered spatial scale. Here, we assess the relative contribution of these factors by studying the population genetic structure of a common avian ectoparasite, the hen flea *Ceratophyllus gallinae*, exploiting two hosts that are sympatric in our study population, the great tit *Parus major* and the collared flycatcher *Ficedula albicollis*. Previous experimental studies have indicated that the hen flea is both locally maladapted to great tit populations and composed of subpopulations specialized on the two host species, suggesting limited parasite dispersal in space and among hosts, and a potential interaction between these two structuring factors.

Results: *C. gallinae* fleas were sampled from old nests of the two passerine species in three replicate wood patches and were genotyped at microsatellite markers to assess population genetic structure at different scales (among individuals within a nest, among nests and between host species within a patch and among patches). As expected, significant structure was found at all spatial scales and between host species, supporting the hypothesis of limited dispersal in this parasite. Clustering analyses and estimates of relatedness further suggested that inbreeding regularly occurs within nests. Patterns of isolation by distance within wood patches indicated that flea dispersal likely occurs in a stepwise manner among neighboring nests. From these data, we estimated that gene flow in the hen flea is approximately half that previously described for its great tit hosts.

Conclusion: Our results fall in line with predictions based on observed patterns of adaptation in this host-parasite system, suggesting that parasite dispersal is limited and impacts its adaptive potential with respect to its hosts. More generally, this study sheds light on the complex interaction between parasite gene flow, local adaptation and host specialization within a single host-parasite system.

López-Pérez, A.M., Gage, K., Rubio, A.V., Montenieri, J., Orozco, L., Suzan, G.(2018) Drivers of flea (Siphonaptera) community structure in sympatric wild carnivores in northwestern Mexico. Journal of Vector Ecology, 43 (1), pp. 15-25.

Host identity, habitat type, season, and interspecific interactions were investigated as determinants of the community structure of fleas on wild carnivores in northwestern Mexico. A total of 540 fleas belonging to seven species was collected from 64 wild carnivores belonging to eight species. We found that the abundances of some flea species are explained by season and host identity. *Pulex irritans* and *Echidnophaga gallinacea* abundances were significantly higher in spring than in fall season. Flea communities on carnivore hosts revealed three clusters with a high degree of similarity within each group that was explained by the flea dominance of *E. gallinacea*, *P. simulans*, and *P. irritans* across host identity. Flea abundances did not differ statistically among habitat types. Finally, we found a negative correlation between the abundances of three flea species within wild carnivore hosts. Individual hosts with high loads of *P. simulans* males usually had significantly lower loads of *P. irritans* males or tend to have lower loads of *E. gallinacea* fleas and vice-versa. Additionally, the logistic regression model showed that the presence of *P. simulans* males is more likely to occur in wild carnivore hosts in which *P. irritans* males are absent and vice-versa. These results suggest that there is an apparent competitive exclusion among fleas on wild carnivores. The study of flea community structure on wild carnivores is important to identify the potential flea vectors for infectious diseases and provide information needed to design programs for human health and wildlife conservation.

Medvedev, S.G. (2018) Morphological Diversity of the Skeletal Structures and Problems of Classification of Fleas (Siphonaptera). Part 6. Entomological Review, 98 (1), pp. 10-20.

The structure of 76 skeletal elements of adult fleas was analyzed, and the distribution of 114 characters with 446 character states over the body tagmata, segments, and morphofunctional complexes was investigated. Among them, 40% of the characters (40) and their states (163) describe the diversity of the structures of the frontal complex (including those of the head and prothorax), which is related to the specific features of flea parasitism. A large part of the characters (18) and their states (83) describe the structures of the nototrochanteral complex of the meso- and metathorax responsible for jumping. The total number of all types of homoplasies (258 states) is almost 1.8 times as great as the number of the states (145) that may be regarded as synapomorphies. The ancestral states (43) comprise a smaller portion of the total number. The proportion of the synapomorphic and homoplastic character states varies between the morphofunctional complexes.

Pathology and Control

Carithers, D., Dryden, M., Everett, W.R., Gross, S., Crawford, J.(2018) Assessment of Frontline® Plus efficacy at 24-hour counts against Tampa 2014 isolate *Ctenocephalides felis* flea infestations on cats and dogs on days 1, 7, 14, 21, and 28. International Journal of Applied Research in Veterinary Medicine, 16 (1), pp. 52-58.

Two studies were performed to determine the 24-hour efficacy of FRONTLINE® Plus (fipronil/(S)-methoprene) against the Tampa 2014 isolate of *Ctenocephalides felis* fleas. This isolate of fleas was collected during a field study where there was a perceived lack of efficacy of FRONTLINE Plus for

Dogs against *Ctenocephalides felis* fleas. Two well-controlled laboratory studies were performed spatially and temporally separate, in two different facilities, against this isolate. In the first study, eight cats were treated with FRONTLINE Plus for Cats on Day 0, and eight cats remained as mineral oil-treated controls. Cats were infested with 100 fleas of the Tampa 2014 isolate on Days 1, 7, 14, 21, and 28, and the fleas were removed and counted 24 hours later. In the second study at a separate facility, 8 dogs were treated with the appropriate dose of FRONTLINE Plus for Dogs for their weight on Day 0, and 8 dogs remained as mineral oil-treated controls. Dogs were then infested with 100 fleas of the isolate on Days 1, 7, 14, 21, and 28, and the fleas were removed and counted 24 hours later. For Study 1, cats treated with FRONTLINE Plus had significantly ($p<0.01$) fewer live fleas than the controls 24 hours post-infestation on each assessment day. Efficacy for FRONTLINE Plus for Cats was 90.3%, 99.6%, 98.8%, 93.3%, and 85.6% on Days 2, 8, 15, 22, and 29, respectively. For Study 2, dogs treated with FRONTLINE Plus also had significantly ($p<0.01$) fewer live fleas than the controls 24 hours post-infestation on each assessment day. Efficacy for FRONTLINE Plus for Dogs was 99.6%, 100%, 100%, 100%, and 97.6% on Days 2, 8, 15, 22, and 29, respectively. FRONTLINE Plus for Cats and FRONTLINE Plus for Dogs demonstrated high efficacy against the Tampa 2014 isolate of *Ctenocephalides felis* fleas throughout the two 29-day studies. It was later discovered that, in the original field study, the owners of the homes in which the perceived inactivity against fleas had taken place had been bathing their dogs in oil-cutting shampoos and/or dish detergent during the study. This is one of many potential confounders that can affect the performance of a topical flea control product. The results of the present studies demonstrate that the true effectiveness of FRONTLINE Plus against *Ctenocephalides felis* fleas was consistent with previous well-controlled studies over the last 20 years. These results indicate that, when there appears to be lack of effectiveness with a flea control product, one should always consider and explore all possibilities.

Leibler, J.H., Robb, K., Joh, E., Gaeta, J.M., Rosenbaum, M.(2018) Self-reported animal and ectoparasite exposure among urban homeless people. Journal of Health Care for the Poor and Underserved, 29 (2), pp. 664-675.

Homeless people in the United States may experience poor hygiene and spend extended periods of time outdoors, which increases exposure to animal and insect vectors of disease. Despite these risks, efforts to understand frequency and risk factors for zoonotic and vector- borne infections among homeless people have been limited. We queried homeless people in Boston, Massachusetts (n=194) to evaluate exposure to urban wildlife and ectoparasites associated with infection. Thirty percent of participants reported seeing rodents daily, and 25% reported daily sightings of cats. Body lice and fleas were reported by 4% and 11% of participants, respectively. Sleeping outdoors and heavy drinking were positively associated with rodent and ectoparasite exposure. Frequent sightings of rodents and rodent feces among homeless people in particular areas may indicate human exposure risk to urban rodent-borne pathogens, including *Leptospira* spp, Seoul hantavirus, and *Rickettsia akari*. Epidemiologic studies of zoonotic and vector-borne infections in this population are warranted.

Uhart, M.M., Gallo, L., Quintana, F. (2018) Review of diseases (pathogen isolation, direct recovery and antibodies) in albatrosses and large petrels worldwide. Bird Conservation International, 28 (2), pp. 169-196.

Albatrosses (Diomedeidae) and large petrels (*Macronectes* and *Procellaria* spp.) are among the world's most rapidly declining birds. Some of the most endangered species, Amsterdam Albatross *Diomedea amsterdamensis*, Indian Yellow-nosed Albatross *Thalassarche carteri* and Sooty Albatross *Phoebetria fusca*, are at risk from recurrent avian cholera outbreaks. Yet little is known about the overall impact of disease in this group. We compiled all available information on pathogens described in albatrosses and large petrel species listed under the Agreement on the Conservation of Albatrosses and Petrels (ACAP) (n = 31). Available reports (n = 53) comprise nearly 60% of ACAP species (18/31). However, only 38% of them focus on threatened species (20/53), and 43% solely report macroparasite findings (23/53). Black-browed Albatross *Thalassarche melanophrys* (Near Threatened) and Southern Giant Petrel *Macronectes giganteus* (Least Concern) are the two species with higher number of publications (29/53, 55% of all papers). Conversely, seven species on the IUCN Red List have three papers or less each. Most existing research has resulted from disease or mortality investigations and baseline studies (28 and 32%, respectively). Pathogens reported in the subset of ACAP species, included bacteria in seven species (39%), viruses in five (28%), protozoa in four (22%), helminths in nine (50%), ectoparasites in 13 (72%) and fungi in one species (5%). Avian cholera, caused by the bacterium *Pasteurella multocida*, appears as the most severe threat to ACAP species. Infections by poxvirus are the most common viral finding, yet entail lower population level impact. Few serosurveys report pathogen exposure in these species, but add valuable baseline information. There are numerous obvious gaps in species and geographical coverage and likely under-reporting due to remoteness, accessibility and sporadic monitoring. This insufficient knowledge may be hampering effective protection and management of populations at risk. Attention to species currently affected by avian cholera is of utmost priority.

[*Glaciopsyllus antarcticus*, *Parapsyllus longicornis*]

Microbial Symbiotes

El Hamzaoui, B., Laroche, M., Almeras, L., Bérenger, J.-M., Raoult, D., Parola, P.(2018) Detection of *Bartonella* spp. in fleas by MALDI-TOF MS. *PLoS Neglected Tropical Diseases*, 12 (2), art. no. e0006189.

Background: Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry (MALDI-TOF MS) has recently emerged in the field of entomology as a promising method for the identification of arthropods and the detection of associated pathogens.

Methodology/Principal findings: An experimental model of *Ctenocephalides felis* (cat fleas) infected with *Bartonella quintana* and *Bartonella henselae* was developed to evaluate the efficacy of MALDI-TOF MS in distinguishing infected from uninfected fleas, and its ability to distinguish fleas infected with *Bartonella quintana* from fleas infected with *Bartonella henselae*. For *B. quintana*, two groups of fleas received three successive blood meals, infected or not. A total of 140 fleas (100 exposed fleas and 40 control fleas) were engorged on human blood, infected or uninfected with *B. quintana*. Regarding the second pathogen, two groups of fleas (200 exposed fleas and 40 control fleas) were fed in the same manner with human blood, infected or not with *Bartonella henselae*. Fleas were dissected longitudinally; one-half was used for assessment of *B. quintana* and *B. henselae* infectious status by real-time PCR, and the second half was subjected to MALDI-TOF MS analysis.

Comparison of MS spectra from infected fleas and uninfected fleas revealed distinct MS profiles. Blind queries against our MALDI-TOF MS arthropod database, upgraded with reference spectra from *B. quintana* and *B. henselae* infected fleas but also non-infected fleas, provided the correct classification for 100% of the different categories of specimens tested on the first model of flea infection with *Bartonella quintana*. As for *Bartonella henselae*, 81% of exposed qPCR-positive fleas, 96% of exposed qPCR-negative fleas and 100% of control fleas were correctly identified on the second model of flea infection. MALDI-TOF MS successfully differentiated *Bartonella* spp.-infected and uninfected fleas and was also able to correctly differentiate fleas infected with *Bartonella quintana* and fleas infected with *Bartonella henselae*. MALDI-TOF MS correctly identified flea species as well as their infectious status, consistent with the results of real-time PCR.

Conclusions/Significance: MALDI-TOF is a promising tool for identification of the infection status of fleas infected with *Bartonella* spp., which allows new possibilities for fast and accurate diagnosis in medical entomology and vector surveillance.

Tomassone, L., Berriatua, E., De Sousa, R., Duscher, G., Mihalca, A.D., Silaghi, C., Sprong, H., Zintl, A. (2018) Neglected vector-borne zoonoses in Europe: Into the wild. *Veterinary Parasitology*, 251, pp.17-26.

Wild vertebrates are involved in the transmission cycles of numerous pathogens. Additionally, they can affect the abundance of arthropod vectors. Urbanization, landscape and climate changes, and the adaptation of vectors and wildlife to human habitats represent complex and evolving scenarios, which affect the interface of vector, wildlife and human populations, frequently with a consequent increase in zoonotic risk. While considerable attention has focused on these interrelations with regard to certain major vector-borne pathogens such as *Borrelia burgdorferi s.l.* and tick-borne encephalitis virus, information regarding many other zoonotic pathogens is more dispersed. In this review, we discuss the possible role of wildlife in the maintenance and spread of some of these neglected zoonoses in Europe. We present case studies on the role of rodents in the cycles of *Bartonella* spp., of wild ungulates in the cycle of *Babesia* spp., and of various wildlife species in the life cycle of *Leishmania infantum*, *Anaplasma phagocytophilum* and *Rickettsia* spp. These examples highlight the usefulness of surveillance strategies focused on neglected zoonotic agents in wildlife as a source of valuable information for health professionals, nature managers and (local) decision-makers. These benefits could be further enhanced by increased collaboration between researchers and stakeholders across Europe and a more harmonised and coordinated approach for data collection.

Population Model for the Bat Flea *Sternopsylla distincta*

(Citation: Bossard, R. L. 2017. Population model for the bat flea *Sternopsylla distincta* (Siphonaptera: Ischnopsyllidae) in temperate caves of the Brazilian free-tailed bat *Tadarida brasiliensis* (Mammalia: Chiroptera). *Journal of the Utah Academy of Sciences, Arts, & Letters* 94: 71-78.)

ABSTRACT: Population dynamics of the bat-flea *Sternopsylla distincta* are not quantitatively known. These fleas parasitize Brazilian free-tailed bats (*Tadarida brasiliensis*) in temperate caves. Using laboratory and field data, I model seasonal population changes. Bat-flea populations are predicted to boom when bats arrive in the spring, and immature-flea stages quickly reach equilibrium densities after a few days. From June to September, all stages are present, with the most abundant being eggs at 90 per m^2 of guano. Larvae equilibrate at 20 per m^2, pupae 1 per m^2, and adults comprise only 10% of the summer's flea population. When bats depart for southern roosts in the fall, bat-flea populations decline, but pupae survive the winter. Results imply that *Sternopsylla distincta* overwinters both as pupae (in northern caves) and as adults (in southern roosts). Fleas are important in many caves as climbing ability and phoresy allow fleas to attain bats on ceilings, creating a "flea-feces loop" of cycling biomass. Fleas reach bats by crawling up cave walls, or when adult-female bats retrieve fallen, flea-infested pups, or through uncommon earwig phoresy. The model provides baseline predictions enabling comparison to field data, and insight into food limitation that appears to regulate bat-flea populations in temperate caves.

Bat fleas are important components of many cave ecosystems. Direct impacts of fleas on bats are not well-studied, but include irritation and blood loss, presumably. Cat flea (*Ctenocephalides felis*) females ingest a blood volume of 13.6 µL per day (Dryden and Gaafar 1991), but no data on bat fleas exists. In addition, it is possible that bat fleas vector [*sic,* transmit] pathogens such as *Bartonella* and *Rickettsia* bacteria, or even white-nose fungus (Dietrich et al. 2016, Lučan et al. 2016).

Bat fleas affect other species in addition to bats. Fleas expel egested bat-blood with flea feces which rain down to the cave floor (where bat fleas mate and lay eggs, whether on the bat or on the cave floor, is a question requiring research). Fleas then attain the ceiling where bats roost. This cycle of material forms a "flea-feces loop" that enriches nutrients in the guano. Guano supports communities unique to caves that require conservation (Gnaspini 2005, Iskali and Zhang 2015).

How bat fleas ascend to the ceiling requires investigation, but the statement "There is no way for the fleas to return to the host once they are removed" (Pape 2014) is incorrect. There are several behaviors that allow bat fleas to move from cave floors, where they develop as immatures, to cave ceilings where they parasitize as adults.

First, adult bat-fleas quest for bats by walking to the highest point available (negative geotropism), then extending their front legs upward (Elbel et al. 1999). The behavior resembles the questing behavior of many ticks and insects. Bats flying through the cave often brush-up against cave obstacles and could pick up questing fleas. Second, when bat pups fall to the floor, mother bats swoop down and pick them up. Fallen bats are infested with large numbers of bat fleas from the cave floor (Elbel et al. 1999). Third, bat fleas cling to the legs of earwigs (Dermaptera), resulting in phoresy

(Hastriter et al. 2017). Fourth, adult fleas are able to walk and climb well (Elbel et al. 1999). Bat-flea adults tend to have exceptionally long legs, but a diminished pleural arch makes them weak jumpers. *Sternopsylla distincta* larvae are unusual, too, having spatulate setae but how these are involved with larval movement is unknown (Elbel and Bossard 2007).

Adult bat-fleas are rare. From the literature, fewer than 15% of bats are typically infested, averaging only one flea per infested bat: "distribution of bat fleas is one of the most enigmatic problems in the entire field.... Doubtless the rarity of bat fleas in this area is in part attributable to lack of collecting. Even in areas that have been thoroughly investigated, however, these fleas are still uncommon" (Lewis 1964).

In this paper, I describe a model of population dynamics of the bat flea *Sternopsylla distincta* in temperate caves containing roosting Brazilian free-tailed bats (*Tadarida brasiliensis*). The bats immigrate into these caves in the spring from southerly roosts, and emigrate in the fall, leaving the fleas in the caves without hosts. The model provides a baseline comparison to field data if it is eventually collected, and offers an insight into factors regulating the numbers of bat-fleas.

Materials and Methods

Flea collection

Bat-flea (*Sternopsylla distincta*) eggs, larvae, pupae, and adults were collected from temperate *Tadarida brasiliensis* nursery caves in the Great Basin Desert and Central Mixed Grassland ecoregions of the United States (Elbel et al. 1999). Larvae were fed guano (Elbel and Bossard 2007). Developmental periods (days) and survivorship (%) of each stage (egg: 5 d, 70%; 1st instar larva: 4 d, 90%; 2nd: 2 d, 60%; 3rd: 7 d, 70%; pupa: 28 d, 50%; unfed adult: 18 d, 30%), and number of eggs each female-adult bat-flea laid per day (one egg per female-day), were estimated from laboratory observations at 20º C (Elbel and Bossard 2007).

Model

The population model is a stage- and age-structured, daily life-table spreadsheet. Each stage is assigned its characteristic developmental period and survivorship from the laboratory estimates.

Model assumptions

1. Bats arrive *en masse* Julian Day 120 (approximately 1 May) and leave Julian Day 240 (approximately 1 September).
2. When bats arrive, they are immediately infested with fleas at a prevalence of 15% and incidence of one flea per bat (numbers estimated from the literature).
3. The bat population remains constant.
4. Any other bat species that overwinter in the cave are not significant refuges for *Sternopsylla distincta*.
5. The entire population of bat fleas overwinters as pupae, that is, there is no contribution of fleas on bats returning to the roost in the spring.
6. Any cocooned flea 28 days old or younger is a pupa, and over 28 days old is a teneral adult (teneral is defined here as an adult that has not left the cocoon). [*sic*, pharate]
7. Teneral adults eclose (open) [*sic*, emerge from] their cocoons whenever bats are present.
8. There is no stochasticity or variance.
9. Temperature in the cave remains constant at 20° C.
10. Mortality happens at the end of each stage.

11. There is no mortality due to food limitation, predation, parasitism, or host-grooming on bat fleas, other than what may be already included in the laboratory stage-specific survivorship estimates.

Results

The bat flea population 'booms' when bats arrive in the spring (Figure 1). Densities of immature stages equilibrate quickly because fleas developing into a stage are balanced by fleas aging out of that stage or dying. The bat-flea population declines after bats depart in the fall. During the winter, pupae survive in dormancy.

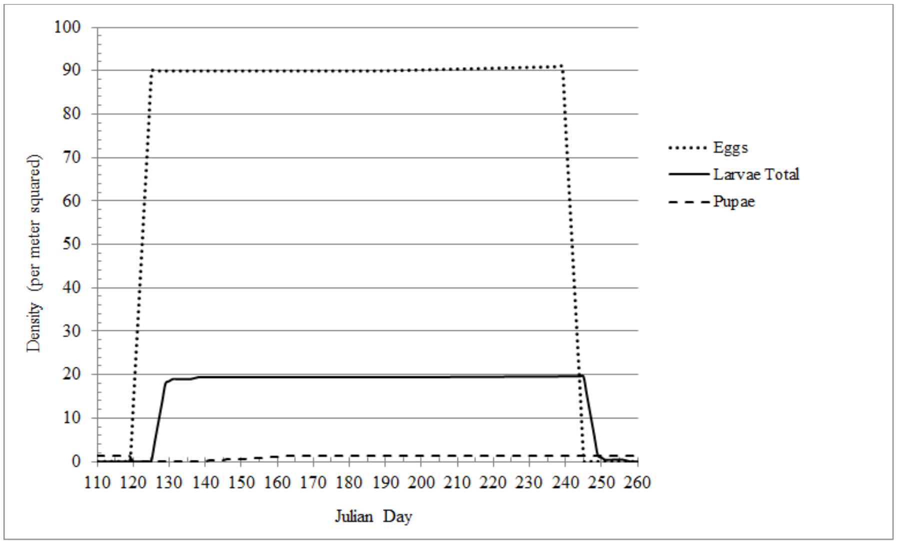

Figure 1. Predicted densities (per m^2) of immature bat-fleas *Sternopsylla distincta* in temperate North American caves during the year (Julian Days). Adult densities (not shown) are lower than pupal densities.

All stages are present and the immature stages in equilibrium from Julian Days 150 to 240 (approximately 1 June to 1 September). The most abundant stage during the summer is flea eggs, equilibrating at 90 per m^2 of guano; larvae equilibrate at 20 per m^2, and pupae at 1 per m^2. Adults comprise only 10% of the equilibrial summer-flea population.

The model predicts that overwintering pupal densities are insufficient to maintain the population.

Discussion

The predicted population dynamics of the model are consistent with observed stage densities, although no quantitative census of immature stages exists, and eggs are too small (millimeter diameter) to be counted easily without a lens (Linley et al. 1994, Elbel et al. 1999). The model does not predict a stable flea population, but nevertheless, fleas are sustained year after year. The discrepancy requires explanation, and further refinement of the model.

First, survivorship in the field may be underestimated by the laboratory data used in the model. Immature bat-fleas probably require nutrients in fresh guano and blood, not the weeks-old guano used to rear the larvae in the laboratory (Elbel and Bossard 2007), although the larvae of many flea species, such as the rat flea *Xenopsylla cheopis*, can survive entirely on organic detritus without the presence of flea feces (Marshall 1981). Second, the assumption that the entire flea population overwinters as pupae may be an over-simplification. At least some *Tadarida brasiliensis* overwinter in dispersed roosts in southern Mexico (Wiederholt et al. 2013), are infested with *Sternopsylla distincta*, and may inoculate northern caves with fleas.

How do adult fleas on bats continuously know when to reproduce? Bat hormones may trigger flea reproduction, but this has never been investigated (Pearce and O'Shea 2007, Lourenço and Palmeirim 2008).

I conclude from the model results that the population dynamics of *Sternopsylla distincta* are complicated, with part of its overwintering population being pupae in northern caves, and part being adults on migrated bats in southern roosts. Dynamics of bat-flea populations are controlled largely by the bats themselves. For the bat-flea *Sternopsylla distincta*, food limitation appears to be the major factor regulating their populations in temperate caves. This is consistent with other cave species. In general, cave ecosystems are food-limited (Poulson 2005).

Future research

1. Compare model predictions to field data.
2. Estimate the contribution that adult bat-fleas carried to the roost by bats arriving in the spring make to the flea population.
3. Extend model to other temperate bat-flea genera in North America (Table 2) (Lewis and Lewis 1994). Comparing *Hormopsylla* (a southern bat-flea) with *Nycteridopsylla* (northern) would be of interest.

Table 1. Niches of the four genera of bat fleas (Ischnopsyllidae) in temperate North America.

Niche characteristics	**Fleas on Bats that Migrate**	**Fleas on Bats that Hibernate**
Colder-environment Fleas	*Sternopsylla*	*Nycteridopsylla*
Warmer-environment Fleas	*Hormopsylla*	*Myodopsylla*

4. Compare temperate versus tropical bat-caves. Southern bat-caves may lack bat fleas, even though many mites are present (Palacios-Vargas, et al. 2011). Mite predation or factors other than food could be limiting bat-flea populations in these caves.
5. Measure nutrient cycling in caves with and without bat fleas.
6. Characterize the physiological state of pupae during the winter.
7. Determine triggers and locations for emergence, movement, mating, and oviposition of adult fleas.

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Fleas in Art and History

The Arabian Nights Entertainments of Antoine Galland

Author: Antoine Galland (Rollot in Picardy, Northern France 4 April 1646 – Paris, 17 February 1715), first European translator of *The Arabian Nights* (excerpt).
Publisher: James Godwin, London, 1813.

When the prince and princess were thus laid together, all the while asleep, there rose a great contest between the genius and the fairy about the preference of beauty. They were some time admiring and comparing them; but at length Danhasch broke silence, and said to Maimoune, You see, and I have already told you, my princess was handsomer than your prince; now, I hope, you are convinced of it.

How! convinced of it! replied Maimoune; I am not convinced of it: and you must be blind, if you cannot see that my prince has the better of the comparison. The princess is fair, I do not deny it; but if you compare them together without prejudice, you will quickly see the difference.

Though I should compare them ever so often, said Danhasch, I could never change my opinion. I saw what I now see at first sight, and time will not be able to make me see more; however, this shall not hinder my yielding to you, charming Maimoune, if you desire it; but I would have you yield to me as a favour! I scorn it, said Maimoune; and I would not receive a favour at such a wicked genius' hands: I refer the matter to an arbitrator; and if you will not consent, I shall get the better by your refusal.

Danhasch, who had ever a great deal of complaisance for Maimoune, immediately consented, which he had no sooner done, but Maimoune stamping with her foot, the earth opened, and out came a hideous, hump-backed, blind, and lame genius, with six horns on his head, and claws on his hands and feet. As soon as he was come out, and the earth had closed up, he, perceiving Maimoune, cast himself at her feet, and then, rising upon one knee, asked what she would please to have with him.

Rise, Cascheasch, said Maimoune; I caused you to come hither to determine a difference between me and that cursed Danhasch there. Look on that bed, and tell me, without partiality, which is the handsomest of those two who lie there asleep, the young man or the young lady.

Cascheasch looked on the prince and princess with great attention, admiration, and surprise; and after he had considered them a good while, without being able to determine which was the handsomest, he turned to Maimoune, and said, Madam, I must needs confess I should deceive you, and betray myself, if I pretended to say one was a whit handsomer than the other: the more I examine them, the more it seems to me each possesses, in a sovereign degree, the beauty which is betwixt them; and if one has not the least defect, how can the other have any advantage? But if either has any thing amiss, it will be better discovered when they are awake, than now they are asleep. Let them then be awaked one after another; and that person who shall express most love for the other by ardour, eagerness, and passion, shall be deemed to have least beauty.

This proposal of Cascheasch's pleased equally both Maimoune and Danhasch. Maimoune then changed herself into a flea, and leaped on the prince's neck, where she stung him so smartly, that he awoke, and put up his hand to the place; but Maimoune skipped away as soon as she had done, and resumed her pristine form; which, like those of the two genii, was invisible, the better to observe what he would do.

In drawing back his hand, the prince chanced to let it fall on that of the princess of China. He opened his eyes, and was exceedingly surprised to find a lady lying by him; nay, a lady of the greatest beauty. He raised his head, and leaned on his elbow, the better to consider her. Her blooming youth, and incomparable beauty, fired him in a moment; of which flame he had never yet been sensible, and from which he had even hitherto guarded himself with the greatest application.

Love seized on his heart in the most lively manner, insomuch that he could not help crying out, What beauty is this! what charms! O my heart! O my soul! In saying which, he kissed her forehead, both her cheeks, and her mouth, with so little caution, that she had certainly been awaked by it, had not she slept sounder than usual through the enchantment of Danhasch.

How, my pretty lady! said the prince, do you not awake at these testimonies of love given you by prince Camaralzaman? Whosoever you are, I would have you to know he is not unworthy of your affection. He was going to awake her at that instant, but refrained himself all of a sudden. Is not this she, said he, whom the sultan my father would have had me marry? He was in the wrong not to let me see her sooner. Had he done so, I should not have offended him by my disobedience, nor would he have had occasion to use me as he has done.

The prince began to repent sincerely of the fault he had committed, and was once more upon the point of awaking the princess of China. It may be, said he within himself, the sultan my father has a mind to surprise me, and has sent this young lady to try if I had really that aversion to marriage which I pretended. Who knows but, having thus laid her in my way, he is hid behind the hangings, to take an opportunity to appear, and make me ashamed of my dissimulation? This second crime would be yet much greater than the first. Upon the whole matter, I will content myself with this ring, which will at any time create in me a remembrance of this dear lady.

He then gently drew off a fine ring the princess had on her finger, and immediately put on one of his own in its place. After this he turned his back, and was not long before he fell into a profounder sleep than before, through the enchantment of the genii.

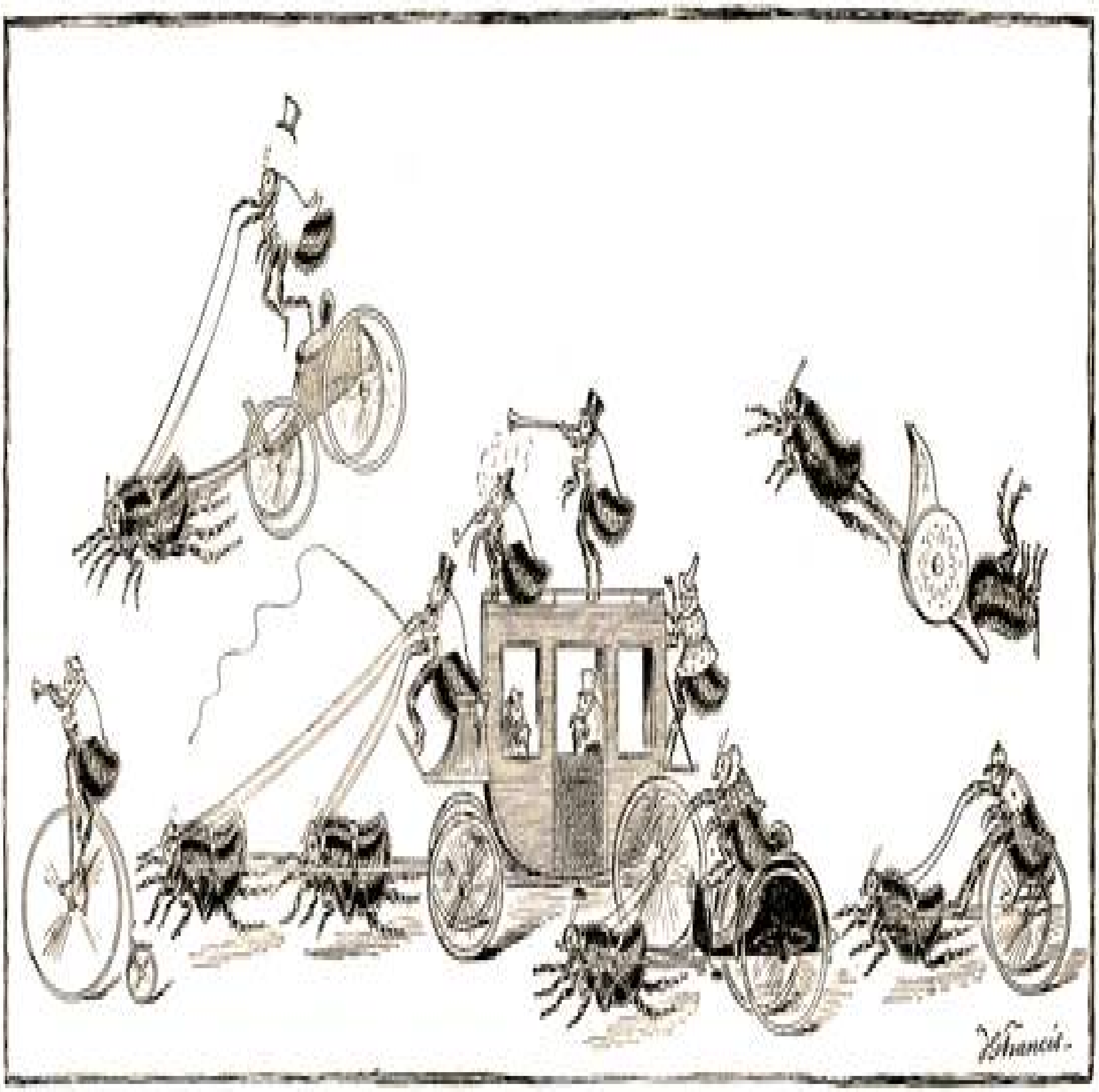
As soon as prince Camaralzaman was sound asleep, Danhasch transformed himself into a flea likewise in his turn, and went and bit the princess so rudely on the lower lip, that she forthwith awoke, started, clapped herself upon her breech, and opening her eyes, was not a little surprised to see a man lying by her. From surprise she proceeded to admiration, and from admiration to real joy, which she conceived at finding him so beautiful and young.

What! cried she, is it you the king my father has designed me for a husband? I am, indeed, most unfortunate at not knowing it before, for then I should not have put my lord and father in a rage, nor been so long deprived of a husband, whom I cannot forbear loving with all my heart. Wake, then, wake, my dear love, proceeded she; for it does not sure become a man that is married, to sleep so soundly the first night of his nuptials.

So saying, she took prince Camaralzaman by the arm, and shook him so violently, as had been enough to have awaked the profoundest sleeper, had not Maimoune at that instant increased his sleep, and augmented his enchantment. She renewed this shaking several times, and finding it did not awake him, she cried out, What is come to thee, my dear! What jealous rival, envying thy happiness and mine, has had recourse to magic, to throw thee into this profound and insurmountable drowsiness; from whence I think thou wilt never recover? Then she snatched his hand, and kissing it eagerly, perceived he had a ring upon his finger which greatly resembled her's, and which she found to be her own. As soon as she saw that she had another on her finger instead of it, she could not comprehend how this exchange could be made; but yet she did not doubt but it was a certain token of their marriage. At length, being tired with her fruitless endeavours to awake the prince, yet well assured that he could not escape her when he awoke, she said, Since I find it is not in my power to awake thee, I will not trouble myself any further about it, but bid thee good night, and then compose myself to rest. At these words, after having given him a hearty kiss on the cheeks and lips, she turned her back, and went again to sleep.

On the next page

"The Go-As-You-Please Race, as seen through a Magnifying Glass",
by J. G. Francis, in Charles Frederick Holder, *St. Nicholas Magazine*, USA, 1886. Engraving.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